

Digital fabrication for stop motion puppets at LAIKA

Henry Brummelman
hbrummelman@laika.com
LAIKA
Hillsboro, OR, United States

John Craney
johncraney@gmail.com
Rubberband LTD
Wellington, New Zealand

Jeff Stringer
jstringer@laika.com
LAIKA
Hillsboro, OR, United States

ABSTRACT

LAIKA integrated digital design methods into its traditional puppet fabrication process and created a faster more iterative and collaborative workflow for building the beautifully bespoke hand-made puppets that come alive in its films.

CCS CONCEPTS

• **Software and its engineering** → **Requirements analysis**; **Software implementation planning**; **Software design techniques**; **Software development process management**.

KEYWORDS

Fabrication, maker, modelling, pipeline, Production, UI/Tools

ACM Reference Format:

Henry Brummelman, John Craney, and Jeff Stringer. 2023. Digital fabrication for stop motion puppets at LAIKA. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595453>

1 INTRODUCTION

LAIKA's films are known for their mix of traditional crafts and advanced technology. Facial shapes for puppets have been designed in CG and 3D printed for replacement animation since *Coraline* in 2008. But the fabrication of the intricately designed puppet bodies has used traditional methods dating back to the shops of Willis O'Brien and Ray Harryhausen.

That process starts with a clay sculpture and involves molding and casting body parts in hard resin or flexible silicone and urethane, manufacturing armatures made of in-house machined joints and swivels, handcrafting "grip blocks" around the metal armatures to give the animator grab points, wigs developed out of tiny synthetic hairs, and custom fabrics specifically made to give the character a costume that moves and functions at scale like real clothing. This process can take months and each step must be done in sequence. Armatures can't start without body molds and costumes can't start without a finished puppet body. Iterations on design require starting again from the beginning. On a film with dozens of original puppets to fabricate this adds up to person-years of work for multiple departments. The integration of digital design tools and 3D printing technology for some parts allowed us to parallelize the

processes and iterate multiple test versions simultaneously so we could innovate with more advanced design ideas and arrive at an approved puppet faster.

2 INTRODUCING DIGITAL METHODS

LAIKA's puppets are works of art and engineering that must perform and be handled for 18 – 24 month shooting schedules. When we began to introduce digital design methods into the traditional fabrication process, it was critical that changes in process or materials did not affect the quality of the final puppet. Our approach was to embrace and respect the traditional artists, and integrate the digital workflow in an already established practical environment.

John Craney and Henry Brummelman led the effort, and focused on the hand-offs between the teams and the dependency they had on one another. Puppet fabrication includes sculpting, armature, molds, casting, costume, and special teams for hands, hair, and paint. In the traditional serialized workflow, each team relies on work product from the team before in order to begin their work. The key element is the body shape under the costume.

After looking at these dependencies, we decided that digital tools could support parallel iteration and collaboration on the body shape by using digitally designed parts that could be duplicated with 3D printing to provide each team with a version of what they need to start work. Revisions could be made based on any team's feedback and redistributed simultaneously.

Using a Zbrush based sculpt instead of scanned clay as a starting point, a DaVinci-posed character is imported into Autodesk Maya and reduced to a basic body shape that is passed to several teams for different uses. The Armature team uses Autodesk Inventor to design metal skeletons to fit inside, and the Costume team uses the same digital model to begin mocking up patterns and draping fabric with CLO 3D over the body shape.

At the same time the new Digifab team designs and prints internal parts for the puppets, such as grip blocks and displacement parts to represent the armatures before they are available. Casting bodies with 3D printed molds and displacements that represent the dimensions of the armature means that duplicate puppets are identical to the original. In hand fabricated process this is never guaranteed, and duplicate builds always need adjustment for costume fits. Using Maya for revisions allows distribution of changes to the Armature, Casting, Mold and Costume teams simultaneously.

The Digifab method serves as the hub where iterations on the body design needed by each contributing team can be managed and published back out to everyone at the same time. Supported by this digital workflow all the assets of each individual department come together repeatedly until there is a final assembled puppet. The body sleeve fits over the machined armature, with range and control points specified by animators, and the full shape is dressed in the final costume.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Talks, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0143-6/23/08.

<https://doi.org/10.1145/3587421.3595453>



Figure 1: Puppet body shape w/ armature design and grip locks.

3 CUSTOM TOOLS FOR DIGI-FABRICATORS

With the introduction of this digital service into the puppet department, digital fabricators also needed an internal workflow and tools.

One of the biggest difficulties in making this transition was handling the volume of iterations and deliveries through Maya in a largely practical department. Henry Brummelman developed a series of tools to automate and standardize operations within Maya so that Jr Modeling artists could produce the volume of changes needed and these tools also made the work accessible to traditional puppet fabricators. It was important to remain flexible and able to respond to all the needs and new ideas quickly. So Henry developed two core tools for digital fabrication that allowed us to handle quick iterations and maintain accuracy and quality.

The Armature System is a custom UI in Maya that combines the initial armature design made in Inventor with standard machined parts used for joints, and allows us to easily exchange parts and rig the armature as we iterate on shapes that affect the range of motion of the puppet.

To revise parts on the spot we developed a Boolean tool that gives the user full control over the entire live Boolean assembly being able to re-use, replace, delete, modify, and duplicate Booleans while giving the user individual control over mechanical allowances that make printed parts fit together.

4 3D PRINTER TECHNOLOGY

Rapid Prototype technology at LAIKA goes back to 2008. We use Stratasys 750 printers for the thousands of replacements faces that are needed for a movie. Those printers are very busy and getting iterative puppet parts in the queue with RP was not efficient.

The Stratasys printer's strength is printing in full color, which due to the parts for puppets being under costume was not needed. So, while we still rely on the Stratasys for many parts, especially larger parts like molds and parts that are not directly used in puppets, we were able to leverage several alternative printer technologies that could be dedicated to puppet part's needs such as printing in strong, flexible, or high-detail material. The Boolean tool allowed us to easily adjust mechanical allowances on parts from different printers so they would fit together.

The Formlabs Resin Printer supports multiple types of resins ranging in resolution, durability, and flexibility. We used them to print high resolution masters, in a very strong material for smaller parts and parts that undergo a lot of force. Formlabs also can use flexible materials that prevent parts from snapping as they control certain shapes of the body.

A 3D Systems wax printer was used for any part with visible skin, like hands and feet and any type of fur. The wax printer has such high resolution that these parts needed very little cleanup work. We also used the wax printer to create casts of metal parts that would otherwise be very hard to machine.

The FDM Raise 3D printer was used for special request like 200% or 500% versions of body assets or supporting practical mold making with larger prints where the precision, duplication and or quality needed for 100% assets was less of a factor.

5 BENEFITS OUTSIDE THE PUPPET DEPARTMENT

The use of digitally modeled assets in puppet fabrication also opened the door to new collaborations with other departments using digital workflows. The Art department could use a rigged version of the puppet body to test interaction and scale for the development of props, vehicles and interior sets.

The costume fabrication team was able to start using CLO 3D to create digital patterns for puppet costumes because they now had digital puppet bodies with armature pivot points to drape material on for testing. The VFX asset team can use the same digital bodies to begin modeling digital doubles with dimensions and control points that match the hand-made versions, and the CLO 3D digital costume patterns worked in Marvelous Designer on the doubles.

The RP department can iterate on neck designs, testing the fit with a digital body shape before committing to a print. This collaboration saves many weeks of testing and revisions that used to only start when the physical puppet body was available. Now teams can work in parallel as designs change.

6 CONCLUSION

Establishing the Digital Fabrication team within the Puppet Department was a critical early step toward LAIKA's goal of building an end-to-end digital pipeline that flows through production design, fabrication, out to the shooting stage and VFX. There are similar efforts in other traditional fabrication teams where we are fully committed to enabling our craftspeople and the traditional way of building puppets and sets but supporting them with digital tools that help them collaborate, and scale what they can do in the shop.

The technology team at LAIKA has future work planned to standardize the tools and connect the teams as these assets move in and out of digital and physical processes. The puppet fabrication team has shown that these methods can speed up iteration, manage revisions and create new ways to visualize physical assets before you build them. Continuing to extend these capabilities will help LAIKA maintain the beautiful hand-made quality of their films while scaling up for multiple productions.